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A CT-based 3D pelvic anatomy dataset of 99 healthy Chinese subjects for anatomical landmark detection

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Accurate automated pelvic anatomical landmark identification for intelligent surgical applications has been limited by the lack of high-quality, standardised 3D data. To address this need, we developed a curated dataset of 99 high-resolution pelvic CT cases from Chinese subjects, comprising CT volumes, segmentation masks, 57 clinically defined anatomical landmark coordinates, pelvic surface meshes (STL files), and reference point clouds. Dataset quality was assessed through annotation completeness, pelvic geometric descriptors, landmark–surface alignment, STL mesh quality checks, and intra- and inter-operator annotation reliability. All cases contained the complete 57-landmark set. The mean distance between the left and right anterior superior iliac spines (ASIS) was 228.5 ± 18.2 mm, and the mean landmark-surface distances were 0.04 mm for cortical landmarks and 1.51 mm for intra-pelvic landmarks. Regeneration of the STL meshes eliminated the detected topological defects with minimal change in enclosed volume. The resulting multimodal dataset provides a reusable resource for developing and evaluating pelvic landmark detection, anatomical measurement, pose estimation, and surgical planning methods.

Background & Summary

Accurate automated three-dimensional (3D) localisation of skeletal anatomical landmarks is essential for image-guided orthopaedic procedures. It provides quantitative guidance for osteotomy planes and implant orientation in joint replacement surgery[1], ensures safe trajectories in spinal pedicle screw fixation[2], and supports stable screw placement in trauma repair, such as femoral neck fracture fixation. Beyond intraoperative applications, precise landmark identification enables population-level anatomical analysis, patient-specific pelvic modelling, personalised implant design, deformity screening, and integration of preoperative planning with intraoperative navigation systems[3]. Despite its importance, current workflows rely predominantly on manual annotation, which is time-consuming, operator-dependent, and difficult to scale[4]. In a study of manual landmark labelling on 3D pelvic models, landmark-dependent discrepancies exceeded 4 mm for some landmarks and reached 10 mm or more for others[5], motivating the development of automated methods.

Existing 3D keypoint detection and point-cloud algorithms are largely general-purpose and designed for generic objects rather than skeletal structures. Generic 3D representation-learning methods and datasets[6,7] do not provide anatomically meaningful keypoint definitions, and unsupervised keypoint-detection approaches such as USIP[8] identify repeatable and stable points on point clouds that are not guaranteed to correspond to

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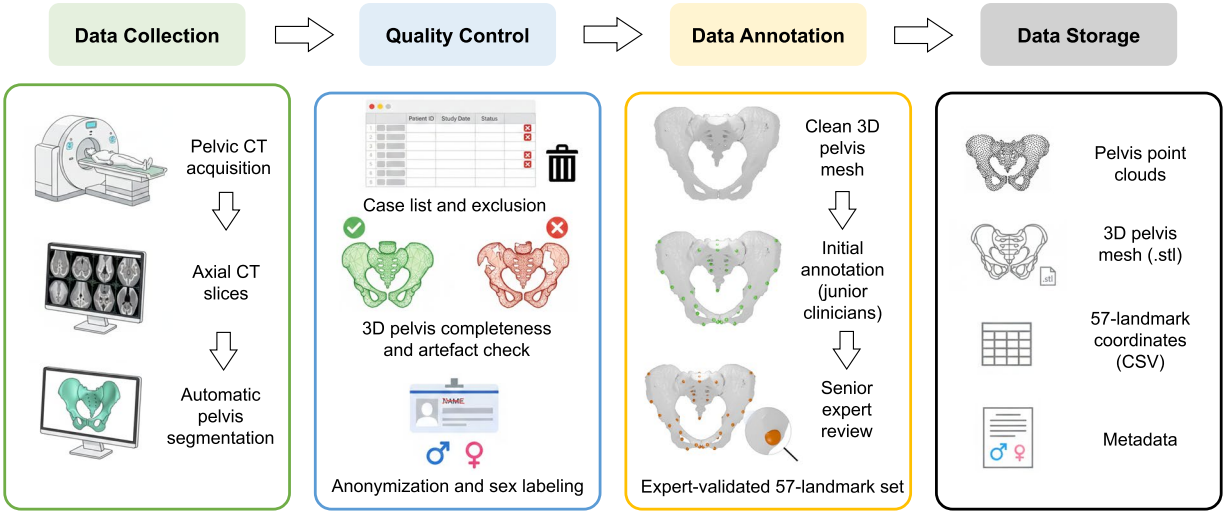


Fig. 1 Overview of the proposed dataset construction. The four-stage process covers data collection, filtering, expert landmark annotation, and storage. The dataset includes 3D pelvis meshes, sampled point clouds, 57 landmark coordinates, and patient-level metadata.

clinically meaningful anatomical structures. In orthopaedics, particularly for the pelvis, these algorithms face additional challenges due to irregular bone shapes, multiple articulating surfaces, and internal foramina, leading to degraded performance. Mainstream point-cloud networks such as PointNet[9] and PointNet ++[10] target generic object classification and segmentation rather than anatomical landmark localisation, and their direct application to clinical pelvic data remains challenging. Traditional 2D landmark detection methods[4,11], are limited to a small number of points with restricted anatomical coverage. Existing 3D pelvic landmark studies often focus on smaller landmark subsets, particular coordinate-system definitions, or task-specific imaging modalities[12,13], and do not provide a unified public release pairing CT volumes, segmentation masks, dense surface representations, and a broad expert-reviewed landmark set. These challenges underscore the need for specialised algorithms that incorporate anatomical priors and are tailored for skeletal and pelvic landmark localisation.

The development of such specialised algorithms requires dedicated datasets. However, existing human landmark datasets are typically limited by small sample sizes, sparse annotations, 2D-only representations[14,15], or anatomical coverage that does not match orthopaedic requirements. Few publicly available resources provide paired de-identified pelvic CT volumes, segmentation masks, STL surface models, dense point clouds, and expert-validated pelvic landmark annotations in a unified data release, leaving a critical gap for algorithm development and benchmarking.

To address this need, we developed a curated pelvic dataset for specialised landmark detection. This dataset contains 99 CT cases reconstructed into high-resolution 3D models, each annotated with 57 expert-validated pelvic landmarks. Annotations followed a standardised protocol with junior and senior clinicians, and dataset quality was confirmed through quantitative metrics covering annotation completeness, landmark–surface alignment, pelvic anthropometry, and mesh quality. This dataset provides a reusable resource for the development and benchmarking of point-cloud learning methods, including PointNet[9], PointNet ++[10], DGCNN[16], PointMLP[17], Point Transformer[18], and LmPT[19]. Its multimodal data, comprising paired CT volumes, segmentation masks, landmark coordinates, surface meshes, and reference point clouds, also enables research on anatomical measurement, pose estimation, patient-specific modelling, and surgical planning.

Methods

The construction of the proposed dataset followed a four-stage pipeline, as illustrated in Fig. 1. First, pelvic CT scans were collected and automatically segmented to generate initial pelvic masks. Second, all cases underwent strict quality control to exclude scans with incomplete pelvic coverage, severe imaging artefacts, or non-standard acquisition protocols. Third, standardised annotation of 57 anatomical landmarks was performed on the reconstructed three-dimensional pelvic meshes by experienced orthopaedic teams, using a two-tier workflow involving junior annotators and senior expert verification. Finally, the curated surface meshes, 50,000-point reference point clouds, landmark coordinates, and patient-level metadata were organised into a structured repository to support public release and downstream algorithm development. Subsequent subsections provide detailed descriptions of the data preparation procedure, landmark definitions, and the annotation workflow.

Data Preparation. Dataset preparation consisted of data acquisition and quality screening. Pelvic CT scans were retrospectively collected from 99 adult patients (59 males, 40 females) who underwent clinically indicated imaging at the Chinese PLA General Hospital (Beijing, China). The cohort represents consecutive eligible cases from the institutional imaging archive during the study period rather than a sex-balanced sample selected for algorithm training. All scans were acquired using a GE MEDICAL SYSTEMS Revolution CT scanner between 7

June 2024 and 29 October 2024, following standardised thin-slice pelvic CT protocols. The in-plane voxel spacing ranged from 0.68 to 1.27 mm (sub-millimetre for 98 of the 99 cases), and the slice thickness was 0.625 or 1.0 mm. In this study, thin-slice CT was defined as having a slice thickness of $\leq$1.0 mm.

The inclusion criteria were:

(i) adult patients with clinically indicated pelvic CT scans;

(ii) complete imaging coverage of the pelvis, including both hip bones and the sacrum;

(iii) slice thickness of $\leq$1.0 mm, suitable for three-dimensional reconstruction;

(iv) complete visualisation of the pelvis, with cortical bone boundaries distinguishable in all predefined landmark regions and all 57 predefined anatomical landmarks identifiable using multiplanar CT and 3D surface views.

The exclusion criteria were:

(i) incomplete pelvic coverage;

(ii) slice thickness $>$1.0 mm or acquisition parameters incompatible with the reconstruction workflow;

(iii) severe motion, metal, or reconstruction artefacts;

(iv) prior pelvic surgery, implants, or gross deformity obscuring the standard pelvic anatomy;

(v) duplicated or non-anonymisable records.

For quality control, all DICOM images were systematically reviewed based on the above inclusion and exclusion criteria, and patient sex was recorded. Where available in the source clinical records, patient height and weight were recorded and are included in the released patient-level metadata (height: 37/99 cases; weight: 36/99 cases). No reliable case-level age field was available in the release sources; the age_years column is retained in the metadata schema but is blank for all cases. At the time of case selection, experienced orthopaedic clinicians reviewed the clinical imaging and available medical records and considered all 99 subjects to have healthy hip and pelvic anatomy, with no evident hip or pelvic pathology.

Landmark definition and annotation. To support surgical planning and related computational tasks[5,20], the dataset defines 57 pelvic anatomical landmarks that collectively capture the essential geometry of the pelvis. Detailed landmark definitions and anatomical region coverage are provided in Supplementary Table S1, following standardised anatomical definitions from Gray's Anatomy[21]: the landmark set covers 19 landmarks on each hemipelvis (38 bilateral landmarks in total), plus 19 sacral landmarks. Landmark annotation was performed using 3D Slicer.

As shown in Fig. 2, the landmark set covers major pelvic regions, including the anterior and posterior iliac spines, ischial tuberosities, the acetabular rim, and internal structures within the sacral and acetabular regions, such as foramina. This comprehensive yet compact landmark configuration provides a unified anatomical reference that facilitates downstream applications including surgical planning, biomechanical analysis, and algorithm benchmarking.

Data Quality Control. A robust quality control framework was implemented using a two-tier approach: initial labelling by junior annotators followed by verification by a senior orthopaedic review team. Junior annotators received systematic training covering the definitions of all 57 anatomical landmarks, 3D Slicer operational standards, and annotation protocols. Initial landmark placement adhered to the principle of "mask cleanup before point placement," with special attention to anatomically challenging landmarks and detailed documentation of the underlying anatomical rationale.

Verification was performed by senior experts with over 10 years of clinical experience in pelvic surgery and more than 5 years of image annotation expertise. The process combined a full review with targeted audits, providing additional scrutiny for landmarks that are difficult to visualise or prone to ambiguity. Challenging points were cross-validated using multiplanar reconstructions, 3D meshes, and original CT scans. Each landmark was classified as qualified, corrected, or rework required. Additionally, 10% of cases were randomly selected for blinded secondary review. Deviations from predefined qualitative review criteria triggered group consultations and iterative refinement of the annotation protocol.

Annotation Workflow and Platform. A two-tier annotation mechanism was implemented using 3D Slicer, and the 3D Slicer annotation workflow is summarised in Fig. 3. The complete dataset-construction workflow consisted of seven steps, including an additional Python-based surface-reconstruction and mesh-quality-control step that is not shown in Fig. 3.

- **CT Data Import:** Qualified pelvic CT volumes were imported into 3D Slicer, with format adaptation and preprocessing.
- **AI-assisted Pelvic Segmentation:** The MONAIAuto3DSeg "Hip and Spine" model automatically segmented pelvic structures, using the default model settings distributed with the 3D Slicer extension.
- **Segmentation Selection and Multi-label Assembly:** The left hip bone, right hip bone, and sacrum were retained as separate anatomical labels within a single multi-label whole-pelvis mask. Misassigned voxels or components were manually reassigned to the correct anatomical label during cleanup.
- **Manual Cleanup of Artefacts:** Junior annotators removed false connections around the sacroiliac joints and pubic symphysis, deleted small isolated artefacts, and used the island-separation tools in 3D Slicer to isolate any remaining disconnected fragments.
- **Surface Reconstruction**: Cleaned segmentation masks were converted to triangular surface meshes using the Lewiner marching-cubes algorithm implemented in scikit-image (version 0.26.0), with degenerate triangles disabled during extraction. Mesh vertices were transformed into physical coordinates using the NIfTI affine and represented in the patient-level LPS convention used for the released STL, NPY, and CSV coordinates.

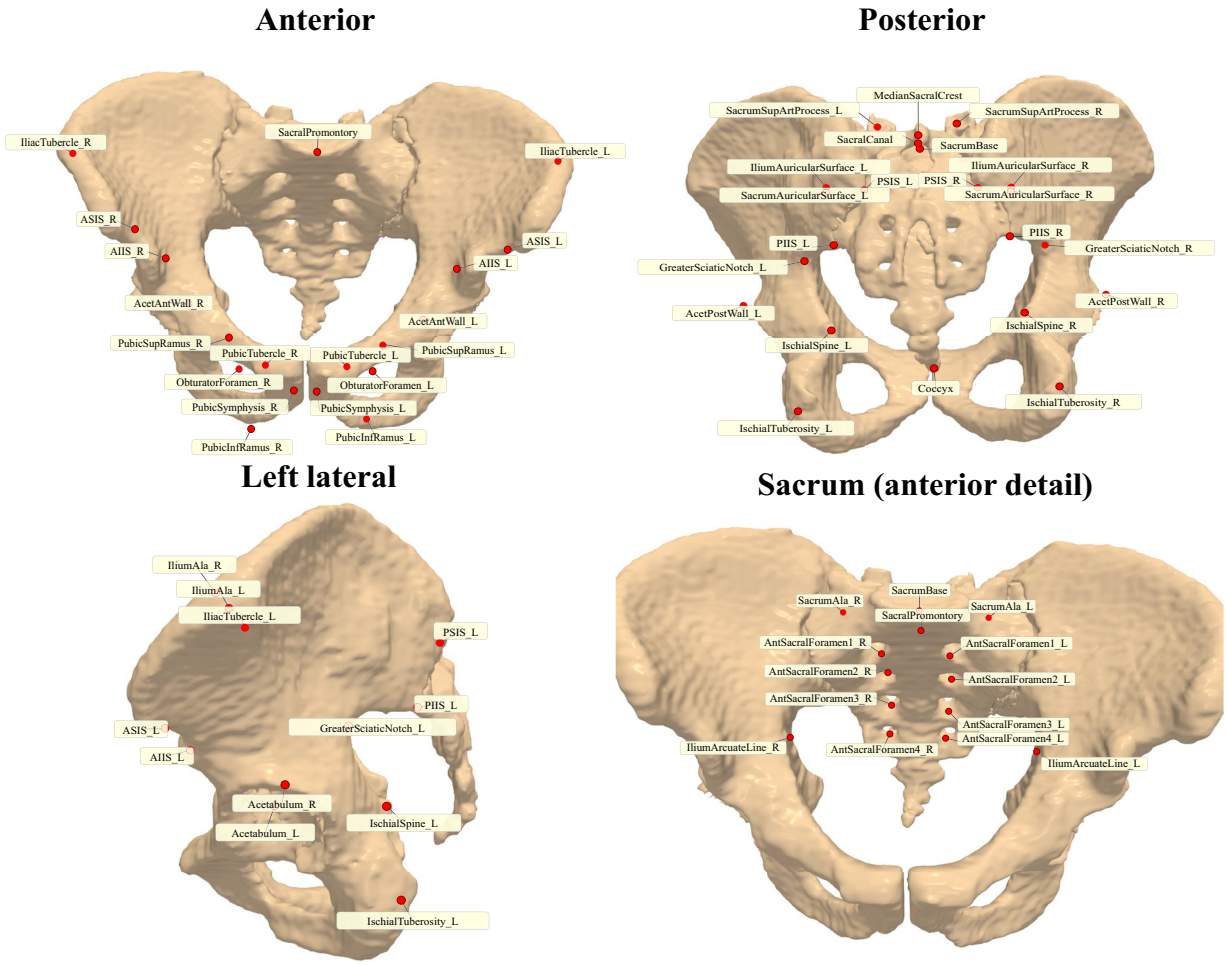


Fig. 2 Comprehensive visualisation of all 57 pelvic landmarks from anterior, posterior, left lateral, and sacral anterior-detail views.

Only zero-volume fragments were removed; no smoothing, decimation, or geometry-altering remeshing was applied. Geometric and topological integrity was audited using trimesh (version 4.12.2), including watertightness, winding consistency, boundary and non-manifold edges, duplicate and degenerate faces, and connected components. Self-intersecting faces were screened using PyMeshLab (version 2025.7.post1). No geometry-altering repair was applied to the final released meshes.

- **Landmark Placement on 3D Mesh:** Landmarks were placed on the cleaned mesh according to the 57-point anatomical definitions using multiplanar reconstructions.
- **Quality Inspection and CSV Export:** Senior experts verified, corrected, and validated landmarks. Coordinates were expressed in millimetres and stored in CSV files with four decimal places for numerical consistency.

Point clouds were generated from the final pelvic meshes using the even surface sampler of the trimesh library with a fixed random seed (0); if the requested number of samples was not returned, the uniform surface sampler was used as a fallback. For public release, we provide a 50,000-point reference point cloud per case in the point_clouds_50000 folder to ensure dense and approximately uniform coverage across anatomical regions. Researchers may further resample or downsample the meshes or reference point clouds to alternative point counts as required by specific algorithms, using the example scripts in the public GitHub repository as a starting point.

All annotations were conducted on Dell Precision 7740 workstations, each equipped with an Intel Core i9-9980HK CPU, 64 GB of RAM, an NVIDIA Quadro RTX 3000 GPU with 6 GB of GDDR6 memory, and an SSD; all workstations ran Windows 10 Pro 64-bit. The software environment comprised 3D Slicer version 5.6.2 with the MONAIAuto3DSeg extension (commit e37d719). The full annotation pass for all 99 patients was completed over approximately five weeks, from 22 October to 26 November 2024, with an average per-case time of approximately 40 minutes covering both manual mask cleanup and 57-landmark placement. Final verification was led by an associate chief physician at the Department of Orthopaedics, the Fourth Medical Center, Chinese PLA General Hospital, with 19 years of experience in pelvic anatomy and medical imaging diagnostics.

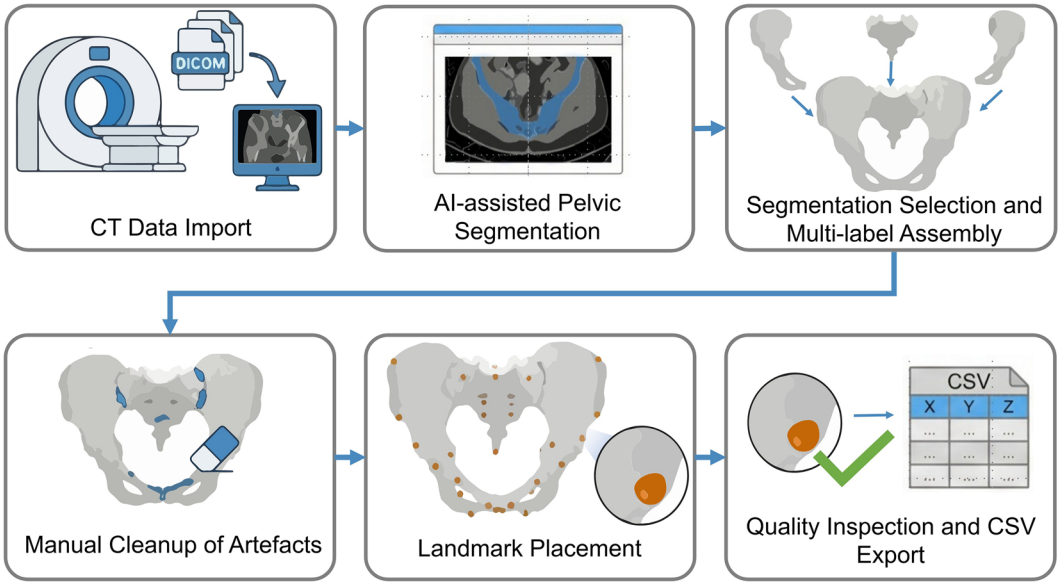


Fig. 3 Annotation workflow. The annotation workflow in 3D Slicer includes importing CT scans, AI-assisted pelvic segmentation, assembling the left hip bone, right hip bone, and sacrum as separate labels within a multi-label segmentation, manual artefact removal, 3D landmark placement, and final quality verification with coordinate export.

Ethics and Privacy. The study was conducted as a retrospective analysis of de-identified pelvic CT scans in accordance with institutional ethical standards and data privacy regulations. All direct patient identifiers, including names, ID numbers, and hospital numbers, were removed prior to data access as part of the de-identification process.

This study was approved by the Ethics Committee of the Chinese PLA General Hospital (approval No. S2020-114-01, titled "Establishment of a Human Skeletal Structure Imaging Database"). As the study involved only retrospective, fully de-identified imaging data and posed no additional risk to patients, the Ethics Committee granted a waiver of informed consent. All released data were de-identified before public deposition.

The released dataset contains volumetric images, segmentation masks, derived pelvic surface meshes, 50,000-point reference clouds, anatomical landmark coordinates, and patient-level metadata. All explicit XYZ coordinates in the released STL, NPY, and CSV files are expressed in millimetres in the patient-level LPS convention. The NIfTI volumes and masks retain their affine matrices for voxel-to-physical coordinate mapping. The dataset is intended for research, algorithm development, and benchmarking purposes, and is not intended for direct clinical diagnosis or decision-making.

Data Records

The proposed dataset has been deposited in Zenodo[22] for research purposes. The dataset comprises 99 de-identified pelvic CT cases, with all explicit XYZ coordinates in the released STL, NPY, and CSV files expressed in millimetres in the patient-level LPS convention, while the NIfTI volumes and masks retain their affine matrices for voxel-to-physical coordinate mapping. Each case contains 57 expert-annotated anatomical landmarks together with corresponding 3D pelvic surface models. Cases are indexed by a common anonymised patient ID to maintain correspondence among CT volumes, segmentation masks, STL meshes, point clouds, and landmark coordinates (e.g., 600001.nii.gz, 600001-mask.nii.gz, 600001-Sacrum.stl, 600001-points-50000.npy, and 600001-Table-XYZ.CSV).

The core data components are stored in separate folders: ct_nifti (de-identified CT volumes in NIfTI format), masks_nifti (segmentation masks in NIfTI format), annotations (CSV files with 57 anatomical landmark coordinates stored with four decimal places for numerical consistency), stl_models (de-identified pelvic surface meshes in STL format), and point_clouds_50000 (50,000-point reference pelvic point clouds in NumPy format). In each NIfTI segmentation mask, voxel value 0 denotes background, 1 denotes LeftHipBone, 2 denotes RightHipBone, and 3 denotes Sacrum. STL files are named after the corresponding anatomical component (LeftHipBone, RightHipBone, or Sacrum). Each landmark CSV uses the anatomical short_name column (e.g., ASIS_L) as the primary landmark identifier, with a parallel anatomical_name_en column providing the full English term (e.g., Left anterior superior iliac spine). A repository-level mapping table, landmark_anatomical_mapping.csv, lists for every landmark the anatomical short name, full English and Chinese names, anatomical region, and laterality, ensuring consistent anatomical naming across masks, meshes, and landmark tables. Example scripts are provided to load the data and resample point clouds if alternative point counts or sampling strategies are required.

Additionally, the dataset includes a patient-level metadata table, patient_metadata.csv. The table contains the anonymised case_id, sex, age_years, height_cm, and weight_kg fields for each included case. Sex is available for all 99 cases (59 male and 40 female). No reliable case-level age field was available in the release sources, so age_years is blank for all cases. Adult status was verified from the source clinical records during eligibility

Metric	Value
Number of patients	99
Landmarks per patient	57
Total landmark labels	5,643
Inter-ASIS distance (mm, mean ± SD)	228.5 ± 18.2
Cortical landmark distance to surface (mm; mean)	0.04
Intra-pelvic landmark distance to surface (mm; mean)	1.51

Table 1. Dataset-level validation for the proposed dataset.

Metric	Unit	All (n = 99)	Male (n = 59)	Female (n = 40)
Inter-ASIS distance	mm	228.5 ± 18.2	230.2 ± 19.1	225.9 ± 16.6
Total pelvic bone volume	cm^3	875.5 ± 145.7	956.9 ± 107.2	755.5 ± 106.9

Table 2. Pelvic geometric parameters stratified by sex. Note: Values are presented as mean ± SD. Full descriptive statistics, including minimum, Q1, mean, standard deviation, median, IQR, Q3, and maximum for each metric, are provided in Supplementary Tables S2 and S3.

screening. Height is available for 37/99 cases and weight for 36/99 cases. Missing values are retained as blank cells and no demographic values were imputed. No personally identifiable information is included in the released metadata.

The Zenodo record is provided as a flat collection of file-type archives together with case-level archives. The file-type archives contain the CT volumes, segmentation masks, regenerated STL meshes, point clouds, and landmark CSV files. In addition, 89 case archives cover all 99 cases: 79 archives contain one case, while the final 20 case IDs are paired in 10 two-case archives named cases_< case_id >_< case_id >.zip to remain within Zenodo's 100-file limit. Each case directory contains the CT volume, segmentation mask, three regenerated STL components, 50,000-point reference cloud, landmark CSV, and one-row metadata.csv. The cohort-level patient_metadata.csv and the landmark_anatomical_mapping.csv are provided as standalone tables.

Technical Validation

We validated the proposed dataset to assess the reliability of the anatomical references and the robustness of the dataset for downstream algorithm development. The evaluation was conducted across six dimensions: annotation completeness, anatomical coverage, geometric consistency, landmark–surface alignment, surface-model quality, and annotation reliability.

These checks support the internal consistency, completeness, and technical integrity of the released representations. They do not constitute independent validation of absolute anatomical accuracy.

Annotation Completeness and Anatomical Coverage. Annotation completeness and anatomical coverage were evaluated across all 99 patients. All cases contained the full set of 57 predefined landmarks, corresponding to 100% completeness, as verified using the consolidated landmark table (5,643 entries in Table 1).

To further characterise anatomical coverage, landmarks were grouped into functionally and anatomically meaningful categories, such as the acetabular rim, sacral foramen, and pelvic ring. Landmark counts of 1,881 for the left hemipelvis, 1,881 for the right hemipelvis, and 1,881 for the sacrum demonstrate complete coverage of both hemipelves and the sacral region without systematic omissions.

Geometric Consistency. Geometric consistency was assessed descriptively using the inter-ASIS distance and total pelvic bone volume computed from the final reconstructed STL meshes. The inter-ASIS distance is a direct landmark-based CT measurement, whereas total pelvic bone volume was calculated as the sum of the enclosed volumes of the final left hip bone, right hip bone, and sacrum meshes. Both metrics are reported as cohort-level anatomical summaries. Coordinate-dependent point-cloud bounding-box dimensions were not treated as standardised anatomical dimensions and are omitted from the validation table.

Across the 99 patients, the inter-ASIS distance was 228.5 ± 18.2 mm (Table 2). This value is close to the 227 ± 16 mm reported in a prospective CT study of 200 adults by Mullaji *et al*[23]. and to the 225 ± 20 mm and 227 ± 13 mm sex-specific Asian model values reported by Ahrend *et al.*[24]. Total pelvic bone volume, computed as the sum of the enclosed volumes of the left hip bone, right hip bone, and sacrum meshes, was 875.5 ± 145.7 cm^3. In the reviewer-specified cadaver CT dataset, Fischer[25] reported mean component volumes of 328 ± 56 cm^3 for the right hip bone, 328 ± 55 cm^3 for the left hip bone, and 203 ± 27 cm^3 for the sacrum; the sum of these component means is 859 cm^3. Because the populations, sample types, segmentation protocols, and volume definitions differ, this comparison is descriptive and does not establish formal equivalence.

Sex-stratified descriptive statistics were calculated for the inter-ASIS distance and pelvic bone volume. The inter-ASIS distance was 230.2 ± 19.1 mm in males and 225.9 ± 16.6 mm in females, while bone volume was 956.9 cm^3 in males and 755.5 cm^3 in females. These values are provided as descriptive characteristics of the dataset rather than for statistical comparison between sex groups.

Anatomical region	Landmark count	Intra-operator Mean ± SD (mm)	Intra 95% CI (mm)	Inter-operator Mean ± SD (mm)	Inter 95% CI (mm)
Left hemipelvis	19	3.68 ± 2.01	2.71–4.65	4.93 ± 1.89	4.02–5.84
Right hemipelvis	19	4.03 ± 1.63	3.25–4.82	5.24 ± 1.64	4.45–6.04
Sacrum	19	2.60 ± 2.10	1.59–3.61	3.51 ± 2.21	2.45–4.57
Total (pooled)	57	3.44 ± 1.98	2.91–3.96	4.56 ± 2.04	4.02–5.10

Table 3. Intra- and inter-operator annotation reliability stratified by anatomical region (n = 10 cases; 57 landmarks per case). Euclidean distances were first averaged for each landmark across the 10 repeated-annotation cases. Regional statistics were then calculated across 19 landmark-level means, and pooled statistics across 57 landmark-level means.

Landmark–Surface Alignment. Geometric alignment relative to the reconstructed pelvis geometry was assessed by computing the point-to-surface distance between each landmark and the corresponding mesh, similar to surface-distance-based validation used in statistical shape modelling and segmentation assessment[26]. This analysis was used as a consistency check to identify potential systematic offsets or preprocessing-related misalignment, rather than as an independent measure of clinical anatomical accuracy. Across all 99 subjects, analysis by landmark type showed that landmarks located directly on the reconstructed bone surfaces were closely aligned with the meshes, with a dataset-level mean distance of 0.04 mm (Table 1). In contrast, landmarks defined inside the pelvic ring rather than directly on the cortical surface exhibited a larger but anatomically expected distance to the surface (mean 1.51 mm, Table 1), consistent with their intended anatomical definition. Point-to-surface distance is a widely used consistency metric in the assessment of reconstructed surface models[27].

Surface Model Quality. To address defects identified in the initial STL exports, all 297 pelvic components were regenerated directly from the corresponding segmentation masks. All 297 final meshes were watertight and winding-consistent. No boundary edges, non-manifold edges, duplicate faces, degenerate faces, or self-intersecting faces were detected. In an internal pre-release audit, compared with the initial STL exports, the median absolute volume difference was 0.049% (95th percentile, 0.223%; maximum, 0.498%), indicating that mesh regeneration eliminated the detected topological defects with minimal change in enclosed volume.

Annotation Reliability. To evaluate annotation reliability, a randomly selected subset of 10 cases was used for both intra- and inter-operator reliability assessment. For the intra-operator analysis, the original annotator repeated landmark placement on the same 10 cases. For the inter-operator analysis, the same subset was independently re-annotated by a senior orthopaedic surgeon who was blinded to the original annotations. Each analysis included all 57 landmarks per case, yielding 570 paired landmark observations. The 10-case subset corresponded to approximately 10% of the released cohort and preserved coverage of the left hemipelvis, right hemipelvis, and sacrum while limiting additional expert annotation burden. Annotation error was quantified as the Euclidean distance between corresponding landmarks and reported as mean ± standard deviation with 95% confidence intervals. For regional summaries, Euclidean distances were first averaged for each landmark across the 10 cases; the reported mean, standard deviation, and 95% confidence intervals were then calculated across landmark-level means within each anatomical region. Because coordinate-wise ICC depends on the chosen Cartesian basis, raw x/y/z ICC values were not retained as a reliability endpoint. When the same rigid transformation is applied to both repeated annotation sets, the Euclidean distance between paired landmarks is unchanged; paired Euclidean distance was therefore used for reliability reporting.

The stratified intra- and inter-operator annotation errors are summarised in Table 3. In the 10-case reliability subset comprising 570 paired landmark observations, repeat annotation by the same operator yielded a mean landmark-level Euclidean distance of 3.44 ± 1.98 mm (95% CI: 2.91–3.96 mm). Independent re-annotation by a second operator yielded a mean distance of 4.56 ± 2.04 mm (95% CI: 4.02–5.10 mm). Region-stratified agreement estimates were of similar magnitude across the left hemipelvis, right hemipelvis, and sacrum, with the lowest errors observed for sacral landmarks. Detailed per-landmark intra- and inter-operator error statistics are provided in Supplementary Table S4.

Usage Notes

Data Use and Processing Considerations. The proposed dataset is intended as a benchmark resource for developing and evaluating 3D pelvic landmark detection, pose estimation, and related geometric learning algorithms on 3D meshes and derived point clouds. Typical applications include keypoint detection, geometric representation learning, shape modelling, and registration.

The dataset originates from a single-centre retrospective cohort of Chinese adult patients collected within a fixed time window and represents consecutively eligible clinical cases after imaging quality and anatomical completeness screening. Accordingly, it reflects a relatively homogeneous clinical population and should be interpreted within this demographic and acquisition context.

All explicit XYZ coordinates in the released STL, NPY, and CSV files are expressed in millimetres in the patient-level LPS convention. The NIfTI volumes and masks retain their affine matrices for voxel-to-physical coordinate mapping.

The released dataset includes high-resolution STL surface meshes, a 50,000-point reference cloud sampled approximately evenly over each pelvic surface, and corresponding anatomical landmark annotations. The

reference clouds were generated with trimesh's even surface sampling, falling back to uniform surface sampling where required (see Methods). The STL files can be loaded using standard 3D geometry libraries (e.g., trimesh, VTK, or PyVista), which support direct mesh parsing and downstream conversion to point-based representations.

Point clouds of arbitrary resolution can be generated from the STL surfaces depending on the requirements of downstream applications. In addition to the high-resolution 50,000-point representation provided for geometric analysis and registration tasks, users may perform further downsampling for learning-based methods. Users may optionally construct fixed-size point clouds for model training by uniform surface sampling followed by farthest point sampling (FPS) to ensure spatial coverage while maintaining computational efficiency; such training preprocessing differs from the released 50,000-point reference point clouds, which are generated with trimesh's even surface sampling (see Methods).

For learning-based experiments, users may centre the point clouds and corresponding landmark coordinates at the per-subject centroid and scale them using a single isotropic per-subject scale factor derived from the maximum Euclidean distance to the centroid. This unit-sphere-style normalisation preserves relative geometry while enabling stable training across subjects with different pelvic sizes; it removes absolute pelvic size information and should not be used for anatomical measurements.

Research Scope and Generalisation. The dataset is intended for research purposes, including algorithm development and benchmarking of geometric learning methods. It is not intended for clinical diagnosis or decision-making. Users should consider that model performance and generalisation may vary when the dataset is applied to populations or imaging protocols beyond the acquisition domain, particularly in multi-ethnic or multi-centre settings.

Limitations and Considerations. Several limitations should be considered when using MedPelvis3D. Metadata are intentionally incomplete: sex is available for all 99 cases, age_years is blank for all cases because no reliable case-level age field was available in the release sources, height is available for 37/99 cases, and weight for 36/99 cases. Missing demographic values are represented by blank cells and were not imputed. The segmentation masks were generated using AI-assisted segmentation followed by manual cleanup but were not independently validated against a reference standard, and no phantom-based validation was performed to establish absolute geometric accuracy. Because the dataset is generated through a multi-step pipeline involving CT acquisition, segmentation, manual correction, surface reconstruction, mesh processing, and landmark annotation, errors introduced at individual stages may propagate or accumulate in the final data. Although the final meshes passed the reported geometric and topological checks, mask-derived marching-cubes surfaces remain dependent on CT voxel resolution and segmentation accuracy. Multiple closed connected components may also remain where spatially separate labelled regions are present; these checks therefore do not constitute independent validation of anatomical accuracy. The cohort consists of clinically indicated imaging cases from a single centre without external or multi-centre validation, the released point clouds were sampled from the reconstructed surface meshes rather than measured directly, and repeated annotation was limited to 10 cases (~10% of the cohort). These factors should be considered when using the dataset for quantitative analysis, model training, or benchmarking.

Data availability

The MedPelvis3D dataset, including the de-identified CT volumes, segmentation masks, 297 regenerated STL meshes, 50,000-point reference clouds, anatomical landmark tables, and patient-level metadata, is publicly available in Zenodo under ref. [22] at https://doi.org/10.5281/zenodo.20473746. The deposited release provides both file-type archives and case-level access covering all 99 cases

Code availability

Custom Python scripts used to reconstruct the STL meshes from the labelled NIfTI masks (rebuild_stl_from_masks.py), audit mesh topology and self-intersections (mesh_qc_and_repair.py), compute geometric and annotation-agreement statistics (compute_geometry_stats.py, compute_annotation_reliability.py), generate surface-sampled point clouds (resample_pointcloud.py), and package the case-level archives (package_cases.py) are available at https://github.com/SweetDeathh/MedPelvis3D. Dependencies and the parameters used for the released dataset are documented in the repository. The analysis-code version corresponding to this submission is tagged as v1.1.2, commit d606872; subsequent documentation-only clarifications are maintained on the repository's main branch.

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Author contributions

N.G. conceived and designed the study, supervised the project, and wrote and revised the manuscript. J.P. developed the experimental code, performed all experiments, and contributed to manuscript writing and revision. G.Z. and J.W. performed dataset annotation. X.S. defined the anatomical landmark criteria and validated the annotated data, with Y.Q. contributing to the development of the landmark definitions. M.L. and Q.Z. provided overall research guidance. M.H. collected and organised the data, developed the annotation and validation protocols, verified the dataset, and participated in manuscript revision.

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Competing interests

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Additional information

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